

Ayrton Parkinson

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Work Experience

Software Engineer, Fullscript

Jan 2025 – Present

- Owned and delivered features end-to-end in a large Ruby on Rails codebase, translating high-level product requirements into technical designs and production-ready implementations
- Led development of a user bulk upload system, replacing synchronous, timeout-prone imports with scalable asynchronous job processing; reduced user onboarding time from hours to minutes
- Architected and delivered customer analytics dashboards, designing and extending GraphQL APIs and building a cron-driven background job pipeline to extract data from Snowflake and support analytics at scale
- Improved data safety and compliance by automating de-identification of PII across multiple domains, safely rolling out production automation that has de-identified 10,000+ accounts

Quantitative Analyst, Canada Deposit Insurance Corporation

Oct 2020 – Dec 2024

- Designed and implemented Python services for loan portfolio valuation, eliminating dependence on external vendors
- Optimized risk assessment workflow by leading the development of comprehensive data dashboards
- Led research to provide senior management with data-driven and actionable advice to manage credit risk

Quantitative Research Analyst, Department of Finance Canada

Oct 2017 – Oct 2020

- Developed statistical models using Python to analyze the impact of financial policy to support senior management
- Automated data analysis pipelines using Python and Excel to improve research efficiency and result reproducibility
- Summarized model outputs and research into concise technical briefings for senior leadership and committee reviews

Projects

AI-Powered Chess Game Analysis and Review Tool

<https://github.com/ayrt-n/chess-review>

- Architected and built a full-stack chess game review tool using TypeScript (React), Java (Spring Boot), PostgreSQL, and RabbitMQ, delivering move-by-move evaluations and feedback
- Built a lightweight Python microservice to expose the Stockfish chess engine over TCP
- Used Vertex AI (Gemini) to generate move feedback, converting technical engine analysis into plain-language explanations
- Built an interactive React frontend with live analysis updates, animation, and responsive playback controls to visualize and review analyzed chess positions

Image-Based Sudoku Solver

github.com/ayrt-n/python-sudoku-solver

- Built an image-based Sudoku solver with Python to extract puzzles from images and compute solutions
- Reduced solve time by 50%+ compared to brute-force backtracking by implementing constraint propagation algorithm to prune the search space before falling back to depth-first search

Finance-Related Social Media Platform

github.com/ayrt-n/invesddit

- Designed and deployed a finance-focused content aggregation and discussion platform using React and Ruby on Rails
- Implemented a ranking algorithm to surface trending content and sort comments, powering personalized home feeds
- Integrated AWS services for user-generated content storage and Twilio SendGrid for transactional email delivery
- Implemented JWT-based authentication and authorization for managing user sessions and permissions

Skills

Programming Languages: Ruby, Python, Java, TypeScript, JavaScript, SQL, HTML, CSS

Frameworks and Technologies: Ruby on Rails, Spring Boot, GraphQL, React, Tailwind, Git, Amazon Web Services

Education

Carleton University, Bachelor of Science in Computer Science

2024

Queen's University, Master of Arts in Economics

2017

University of Ottawa, Bachelor of Social Science in Economics

2016